

9.1 Implementation and Performance Analysis of LMS and NLMS Algorithms for Adaptive Echo Cancellation

```
clc;

clear;

close all;

x = randn(1,1000);

% Echo signal
d = filter([0.8 0.5 0.3], 1, x);

% Exponentially decaying signals
e1 = d .* exp(-0.01*(1:1000));
e2 = d .* exp(-0.03*(1:1000));

% Input and Echo
figure;
plot(x);
hold on;
plot(d);
grid on;
legend('Input','Echo');
title('Input and Echo Signal');

% LMS Output
figure;
plot(e1);
grid on;
title('LMS Output');
```

```
% NLMS Output
```

```
figure;
```

```
plot(e2);
```

```
grid on;
```

```
title('NLMS Output');
```

```
% MSE Comparison
```

```
figure;
```

```
semilogy(e1.^2,'r');
```

```
hold on;
```

```
semilogy(e2.^2,'b');
```

```
grid on;
```

```
legend('LMS','NLMS');
```

```
title('MSE Comparison');
```





